

Generation tokens — pricing options

AI-MODEL-CHAT · GENERATION STUDIO · DECISION DOCUMENT

Decision document for the Generation Studio billing unit. Nothing here is implemented yet. Read sections 1–3, pick an option in section 4, answer the three open questions in section 9.

1. What is wrong with credits

A photo costs **20 credits**. A five-second clip costs **230**. Starter includes **600 a month**, Pro 2,500, Agency 7,500.

None of those numbers mean anything to a creator. The scale exists because `CREDIT_COST_USD = 0.002` made one credit a fixed slice of *provider cost* — excellent for margin discipline, useless as a thing a person reads before pressing Generate. A creator cannot tell whether 230 is a lot.

Two changes:

1. **Re-denominate to tokens at human scale.** One token is about one photo. A clip is about twelve. A month's allowance is a two- or three-digit number.
2. **Cut the retail price from ~15x provider cost to 2.3–3.8x**, on a ladder that discounts 40% by volume. A photo is **€0.15** at the smallest pack and **€0.09** at the largest — at or below every comparable platform.

The margin discipline is kept exactly: a token is still a fixed slice of provider cost, the slice is just twenty times bigger.

2. What the market charges

Platform	Entry plan	Unit	Per image	Per 5s clip
Higgsfield	\$19 → 270 credits	credits	~2 cr (~ \$0.14)	7–29 cr
Candy AI	\$12.99/mo + token packs	tokens	~2 tk (~ \$0.20)	5–10 tk
Kling	\$8.80 → 660 credits	credits, expire monthly	—	—
Runway	~\$15 → 625 credits	credits	—	Max \$95 → 9,500 cr
Leonardo	\$12 → 8,500 tokens	tokens	~\$0.035	—
SeaArt	\$5.99	"stamina"	~12 cr/query	—
Supercreator	\$99/account	+5% of AI-driven sales	—	—
Botly	~\$129	+ usage fees	—	—
Us, proposed	€49 → 30 tokens	tokens, expire monthly	€0.15	€1.80
Us, today	€49 → 600 credits	credits	~€0.56	~€6.40

Three things fall out of this:

The market has already split into low-number and high-number units. The platforms creators actually enjoy using — Higgsfield, Candy — charge 2 of *something* per image. The ones people complain about — Kling, Runway, and us — charge hundreds. Low numbers are not a cosmetic choice; they are what makes a price legible at the moment of spending.

Leonardo's \$0.035 is not a comparable number. That is SDXL-class output with no identity lock. Our pipeline conditions on an approved vault photo *and* faceswaps from the same photo, on Seedream, which bills us \$0.04 before we add anything. We cannot and should not try to reach \$0.035.

Supercreator prices on a completely different axis: \$99 flat plus 5% of AI-driven net sales. Worth knowing it exists, but it needs attributable revenue per message, which we do not have. Not proposed here.

3. What a token costs us

Unchanged from today, just re-pegged. `TOKEN_COST_USD = 0.04` — what one token is allowed to cost us at the provider. Every generation is priced `ceil(provider_cost / TOKEN_COST_USD)`, so rounding always favours us. Measured provider costs (from live bills, in `credits.PROVIDER_COST_USD`):

	cost
Seedream 4.5 / 5.0 Pro, any size	\$0.04
Nano Banana 2, 2k	\$0.10255
Nano Banana Pro, 2k	\$0.138
Wan 2.5, per second 720p	\$0.09076
Wan 2.7, per second 720p	\$0.10076

Everything else in the table is a deliberate over-estimate. **That mattered little at 15x margin and matters a great deal at 2.3–3.8x** — see section 8.

4. The three options

Option A — flat rate

One token per photo, ten per clip, whatever model you pick. The clearest pricing a person could be given. **It does not work.** At €0.15 a token — the smallest pack's rate — here is what a flat rate actually earns against what it costs:

Generation	Costs us	Flat price	Earns	Multiple	
Photo — Seedream 2k/4k	\$0.040	1 token	\$0.153	3.82x	OK
Photo — Nano Banana 2, 2k	\$0.103	1 token	\$0.153	1.49x	under floor
Photo — Nano Banana Pro, 2k	\$0.138	1 token	\$0.153	1.11x	under floor
Photo — Nano Banana Pro, 4k	\$0.276	1 token	\$0.153	0.55x	loses money
Clip — Wan 2.5, 5s 720p	\$0.454	10 tokens	\$1.530	3.37x	OK
Clip — Wan 2.7, 5s 720p	\$0.504	10 tokens	\$1.530	3.04x	OK
Clip — Wan 2.5, 5s 1080p	\$1.135	10 tokens	\$1.530	1.35x	under floor
Clip — Seedance, 10s 1080p	\$3.000	10 tokens	\$1.530	0.51x	loses money

To clear the 2.25x floor flat we would need a **5-token photo** and a **9-token clip** — at which point it is no longer flat in any useful sense. The only way to keep "1 token = 1 photo" flat is to **remove every image model except Seedream and cap video at 720p**, which throws away Nano Banana Pro and the whole premium tier.

Recorded so it is not re-proposed. At 15x margin this worked; at these prices it cannot.

Option B — graded, 1 token ≈ 1 photo (*recommended*)

A standard photo is 1 token. Better models cost more, in single digits.

Images

Model	2k	4k
Seedream 4.5	1	1
Seedream 5.0 Pro	1	1
Nano Banana 2	3	6
Nano Banana Pro	4	7

Video (whole job rounded up, so any length is priced)

Model	720p 3s	5s	10s	1080p 3s	5s	10s
Wan 2.5	7	12	23	18	29	57
Wan 2.7	8	13	26	19	32	63
Seedance 2.5	9	15	30	23	38	75
Face swap	9	15	30	23	38	75

Audio add-on: **3 tokens** a clip. Upscale and the NSFW check disappear as separate line items — at this scale they round to 1 token, which would be a 10x overcharge on a \$0.004 operation, so they fold into the base price.

Monthly allowance: **10 / 30 / 125 / 375** (Demo / Starter / Pro / Agency).

Option C — graded, 2 tokens = 1 photo

Identical shape, doubled. Matches the Candy AI and Higgsfield convention exactly, so a creator arriving from either reads our prices without conversion.

Model	2k	4k	Model	720p 5s	1080p 5s
Seedream 4.5	2	2	Wan 2.5	23	57
Nano Banana 2	6	11	Wan 2.7	26	63
Nano Banana Pro	7	14	Seedance 2.5	30	75

Audio: 5 tokens. Allowance: **20 / 60 / 250 / 750**.

The case for C is finer grading — the gap between Seedream and Nano Banana Pro is 2 → 7 rather than 1 → 4, so a price difference is visible without being dramatic. The case against is that every number is twice as big for no extra information, and "1 token = 1 photo" is the single sentence that makes the whole system explainable.

Side by side

	A — flat	B — 1 tk/photo	C — 2 tk/photo
Standard photo	1	1	2
Premium photo 4k	1	7	14
5s 720p clip	10	12	23
Pro allowance	125	125	250
Explainable in one line	yes	yes	nearly
Clears the 2.25x margin floor	no	yes	yes
Needs models removed	yes	no	no

5. What a token costs, in euro, dollars and pounds

Euro is the base currency. Prices are set in euro — matching the existing €49 / €149 / €349 plans — and the dollar and pound ladders are hand-set round numbers alongside, not live conversions. A pack never costs €13.47 and never moves because the exchange rate did.

The floor is still checked in dollars, because the providers bill us in dollars. Each euro and pound price is converted at a **conservative** reference rate (€1 = \$1.02, £1 = \$1.18) and must still clear 3.0x. Using a pessimistic rate means an ordinary FX swing cannot quietly push a price under cost.

The ladder — one price for everyone

Tokens	EUR	USD	GBP	€/token	margin	buys
100	€15	\$16	£13	€0.150	3.82x	100 photos / 8 clips
500	€70	\$75	£62	€0.140	3.57x	500 photos / 41 clips
1,000	€130	\$139	£115	€0.130	3.31x	1,000 photos / 83 clips
2,000	€220	\$235	£195	€0.110	2.81x	2,000 photos / 166 clips
5,000	€450	\$479	£395	€0.090	2.29x	5,000 photos / 416 clips

A photo runs **€0.15 down to €0.09** and a five-second clip **€1.80 down to €1.08**, depending on pack size. Generation prices quoted in cash use the smallest pack's rate, because that is the marginal price of buying more — the same reason `credits.credit_rate_usd()` reads `PACK_SIZES[0]` today.

MIN_MARGIN_MULTIPLE moves **4.0** → **2.25**. The two largest packs sell at 2.81x and 2.29x, under the 3.0x the smaller ones clear. That is the volume discount working as intended — gross margin is still 64% and 56% — but the floor is a build-gate assertion that runs at import, so it must come down or nothing starts.

The ladder already is the discount — drop the tier bands

Today Pro and Agency buy credits ~13% and ~27% cheaper than Starter. This ladder already falls 40% from smallest pack to largest, which is the same incentive bought by volume rather than by subscription tier. Stacking the tier bands on top would put the 5,000 pack at roughly **1.7x cost**.

Recommendation: drop the bands. Price on size alone, and let the upgrade incentive live in the included allowance instead — which is what section 6 is about.

6. The included allowance is now the problem

Re-denominating exposes something the old scale was hiding. Here is what each plan actually includes today:

Plan	Price	Credits now	= Tokens	= Photos	= Clips	Costs us
Demo	free	200	10	10	0	\$0.40
Starter	€49	600	30	30	2	\$1.20
Pro	€149	2,500	125	125	10	\$5.00
Agency	€349	7,500	375	375	31	\$15.00

Starter is €49 a month for two video clips. Agency is €349 for thirty-one. For a product whose entire pitch is "generate content your fans will pay for", these allowances are not credible — and at the old €0.56-per-photo pricing they at least *sounded* substantial. At €0.15 they are visibly thin: Pro's 125 tokens are about €19 of value inside a €149 plan.

Raising them is cheap, because generation costs us very little:

Plan	Proposed	= Photos	= Clips	Costs us/mo	% of plan price
Demo	25	25	2	\$1.00	—
Starter	100	100	8	\$4.00	8%
Pro	350	350	29	\$14.00	9%
Agency	1,000	1,000	83	\$40.00	11%

That is a 3x increase in what every plan includes for single-digit percentages of revenue, and it restores the upgrade ladder that dropping the discount bands takes away. **Recommended.**

7. What is not changing

- **Monthly allowance still expires monthly.** No rollover. Bought tokens never expire. Same as Kling and Higgsfield, same as the code does today.
- **Generation stays admin-only.** `/studio` and every `/api/generate/*` route keep `_require_admin` and keep 404-ing rather than 403-ing. Token checkout stays shut too — nobody should buy tokens for a feature that is not offered yet. This document prices the feature; it does not open it.
- **The ledger stays append-only.** Balance is the sum of rows, never a counter, because people will buy these. A spend still drains the expiring allowance before anything purchased; a refund still returns tokens to the bucket they left.
- **Pricing stays pegged to provider cost.** One number, `TOKEN_COST_USD`. A new model is a table entry, not a pricing decision.

8. The risk this creates

Most of the price table is not measured. From `credits.py`'s own notes, the following are deliberate over-estimates carried because *a guess that is too low loses money on every generation and nothing reports it*:

- **Seedance 2.5** — refused the probe at ByteDance's moderation end before billing anything. Entire row is a guess.
- **Every 4k image rung** — priced at twice 2k because the 4k result came back before the provider reported a cost.
- **480p and 1080p video** — only 720p was measured. 480p is charged at the 720p rate; 1080p keeps the old table's 2.5x shape.
- **The swap models** (`p-video-replace`, `ml-face-swap`) — unmeasured.
- **The audio add-on** — \$0.10 assumed, and per `CLAUDE.md` none of the audio model ids or field names have been verified against the live catalogue.

At 15x margin a 2x over-estimate was invisible. **At 2.3x on the largest pack it is the difference between a healthy margin and selling under cost.** Before this ships, measure at minimum: Seedance at 720p, one 4k image on each Google model, and one 1080p clip. Run `imagegen.search_models('audio')` against a live Runware key and correct the audio defaults.

The build gate protects us either way — `_assert_generation_floor()` walks the entire job matrix at import and refuses to start if any rung would sell under cost — but it can only check the numbers it is given.

9. Decisions needed

1. **Which denomination?** A is ruled out by arithmetic. Recommend **B** — one token, one photo, the sentence that makes the system explainable.
2. **Raise the included allowances?** Recommend **yes**: 25 / 100 / 350 / 1,000. Starter at two clips a month is not a sellable entry tier.
3. **Drop the tier discount bands?** Recommend **yes** — the size ladder already discounts 40%, and stacking tier bands on top would reach ~1.7x cost.

Once these are settled the implementation is mechanical: `credits.py` and its tests first (pure, no I/O, the floor assertions prove the numbers before anything else moves), then the ledger, then the routes and checkout, then the studio's own labels.